

Travis Brown

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OBJECTIVE

Seeking internship in Embedded/Firmware Engineering from May 2027 to August 2027.

EDUCATION

Rochester Institute of Technology

Rochester, NY

Bachelor of Science and Master of Science in Computer Science (Accelerated BS/MS Program)

Expected: December 2027

GPA: 3.96

SKILLS

Languages: C, C++, Python, Rust, Java, JavaScript, SQL, MIPS Assembly

Embedded: STM32 / ESP32 microcontrollers, ThreadX RTOS, CAN, SPI, I2C, UART, PWM, ADC, OpenOCD, Logic Analyzer

Tools/Environments: Git, GitHub, GitLab, Linux, macOS

WORK EXPERIENCE

Embedded Software Intern

GE Vernova

May 2026 – Present

- Developed embedded firmware in C for STM32F7 microcontrollers, implementing a full I/O subsystem from scratch for TransNEXT industrial radios with support for digital in/out, analog input, PWM output, and edge detection across three configurable I/O ports.
- Debugged hardware and firmware issues using Saleae logic analyzer and oscilloscope, iterating on target hardware via J-Link Segger.
- Built a Flask-based web application monitoring a dynamic TransNEXT radio network, diagnosing and resolving memory leaks to achieve stable multi-day continuous runtime.

QA Automation Intern

Lockheed Martin

May 2025 – August 2025

- Produced a Python-based automation tool that streamlined static code review, reducing QA review time by 75% (from 20 mins to 5 mins).
- Automated over 60% of manual software validation checks, significantly reducing QA workload and improving team efficiency.
- Built and maintained CI/CD pipelines using GitLab CI, contributing to faster and more reliable testing cycles.

PROJECTS

Real-Time Vehicle Control Unit (VCU)

RIT Electric Vehicle Team

October 2025 – Present

- Engineered embedded firmware in C++ for a custom-designed PCB serving as the central VCU for an electric motorcycle.
- Established CAN and CANopen communication between the VCU, motor controller, power subsystem, battery management, and more.
- Built board-level fault handling, drive-by-wire control, and system health monitoring on an STM32F446 managing dual CAN buses across powertrain and accessory networks.
- Designed CAN message formats and CANopen object dictionary; integrated a Simulink-generated 300 Hz state machine within ThreadX RTOS to meet hard real-time motor controller timing.

Handheld Chip-8 Emulator Console

Personal Project

June 2025 – December 2025

- Developed a Chip-8 emulator in Rust for a Raspberry Pi, with display, input, and sound using GPIO and SPI communication.
- Fabricated a 4x4 matrix keypad PCB in KiCad, including schematic and layout, and manually soldered key switches, caps, and diodes.
- Created a custom SPI display driver for an SSD1309 OLED screen, avoiding external graphics libraries for low-level control.

STM32 F4 Peripheral Support

RIT Electric Vehicle Team

December 2023 – November 2024

- Implemented driver support for PWM, ADC, SPI, and timers on the STM32F4 microcontroller family, working with the STM32 HAL.
- Leveraged STM32's CubeMX and CubeIDE to streamline configuration and integration with EVT's hardware-agnostic API.
- Applied microcontroller datasheets and reference manuals to develop reliable, reusable peripheral drivers.